21 $f(x) = \dfrac{1}{x+2} + \dfrac{1}{x-3}$

(a) Work out f(5)
Give your answer as a fraction.

$f(5) = 1/7 + 1/2 = 2/14 + 7/14 = 9/14.$

1 mark
Substitute x=5

1 mark
9/14

(2)

(b) Write down a value of x for which $f(x)$ is not defined.

$x = 3$ makes the denominator $x - 3$ zero.

1 mark
3 (or −2)

(1)

Given that $f(x) = 4$

(c) find the possible values of x.

Give your answer in the form $\dfrac{p \pm \sqrt{q}}{r}$ where p, q and r are positive integers.

$$\frac{2x - 1}{(x + 2)(x - 3)} = 4$$

1 mark
One fraction

$2x - 1 = 4(x^2 - x - 6)$

1 mark
Multiply and rearrange

$4x^2 - 6x - 23 = 0$

1 mark
Correct quadratic

$$x = \frac{6 \pm \sqrt{404}}{8}$$

1 mark
Quadratic formula

$$x = \frac{3 \pm \sqrt{101}}{4}$$

1 mark
Simplest surd form

(5)

(Total for Question 21 is 8 marks)